



Pranesh S A 251801234-2025-AIDS ▾

**P2**

|              |                                   |
|--------------|-----------------------------------|
| Started on   | Thursday, 8 October 2026, 8:50 AM |
| State        | Finished                          |
| Completed on | Thursday, 8 October 2026, 8:52 AM |
| Time taken   | 1 min 41 secs                     |
| Marks        | 1.00/1.00                         |
| Grade        | 10.00 out of 10.00 (100%)         |

**Question 1** | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

**For example:**

| Input               | Result         |
|---------------------|----------------|
| 5<br>67 34 12 98 78 | 12 34 67 78 98 |

**Answer:**

```

1  #include <stdio.h>
2
3  int main() {
4      int a;
5      if (scanf("%d", &a) != 1 || a <= 0) {
6          return 0;
7      }
8
9      int arr[a];
10     for (int b = 0; b < a; b++) {
11         scanf("%d", &arr[b]);
12     }
13
14     int c[a], d[a], e = 0;
15     c[e] = 0;
16     d[e] = a - 1;
17     e++;
18
19     while (e > 0) {
20         e--;
21         int f = c[e];
22         int g = d[e];
23
24         int h = arr[g];
25         int i = f - 1;
26
27         for (int j = f; j < g; j++) {
28             if (arr[j] <= h) {
29                 i++;
30                 int k = arr[i];
31                 arr[i] = arr[j];
32                 arr[j] = k;
33             }
34         }
35         int k = arr[i + 1];
36         arr[i + 1] = arr[g];
37         arr[g] = k;
38
39         int l = i + 1;
40
41         if (l - 1 > f) {
42             c[e] = f;
43             d[e] = l - 1;
44             e++;
45         }
46
47         if (l + 1 < g) {
48             c[e] = l + 1;
49             d[e] = g;
50             e++;
51         }
52     }

```

|   | Input                               | Expected                      | Got                           |   |
|---|-------------------------------------|-------------------------------|-------------------------------|---|
| ✓ | 5<br>67 34 12 98 78                 | 12 34 67 78 98                | 12 34 67 78 98                | ✓ |
| ✓ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✓ |
| ✓ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90    | 1 2 3 4 5 6 7 8 9 10 11 90    | 1 2 3 4 5 6 7 8 9 10 11 90    | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)